TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	High-efficient hydrogen internal combustion engine for power generation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical materials		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles & Shipbuilding/ Marine Systems	Engines and Power Transmission Systems & Hull/Deck and Propulsion System Components	New and Renewable Energy	Hydrogen
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology* This proposal develops a high-efficient internal combustion engine with high power density running on hydrogen fuel based on an existing heavy duty (diesel) engine platform (1). The resulting engine is zero emission with better performance regarding efficiency, power density and load response compared to (non-sustainable) power generation alternatives. The Korean SME partner is a manufacturer/supplier of power generation equipment e.g. for heavy duty automotive, maritime, agriculture, construction or stationary power generation. He will support the project by supplying base engine hardware and integration of the developed zero emission engine into the prototype final application. Korean partner will convert the developed methodology			

Rationale for Joint R&D Project	into concrete product development steps for integration into own development process. *Describe the core concept, main functions, and potential uses of the proposed technology* For the timely realization of the climate targets, new CO2-free solutions for power generation are needed, next to and complementary to battery-electric solutions. The hydrogen internal combustion engine is a promising zero emission solution. For successful upscaling of hydrogen applications, these applications must have competitive costs and performance. The technology developed in this proposal fulfills these requirements. Depending on the targeted application a selection for the used combustion concept will be made. Here, both SI as well as CI hydrogen ICE concepts are possible. An existing (diesel) engine platform can be used as base engine. The expertise of TNO [see references section 6 (2),(3),(4),(5)] is used to optimize the engine efficiency and reduce OPEX, while realizing competitive load performance. By combining the Korean's SME partner expertise on engine-application engineering with TNO's expertise and facilities on (high pressure) hydrogen combustion optimization and engine performance and validation testing, the project targets a TRL-5 demonstrator in which the prototype zero emission engine is integrated in the target application. The demonstrator will show feasibility and performance of the solution. For the Korean partner it is a crucial step towards extending his portfolio with a sustainable solution for Asian and European market.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 1. Zero emission internal combustion engine 2. High efficiency 3. High power density 4. Based on existing (diesel) engine platform 5.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* The R&D objectives of this project are as follows: 1) Development, application and knowledge transfer of engine conversion methodology from fossil fuel to hydrogen with focus on both hardware and software adaptations 2) Realization of prototype high-efficient hydrogen engine with validated performance 3) Development and application of methodology for prototype engine integration into target application			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* The R&D target specifications are as follows: 1) Engine efficiency exceeding conventional diesel engine efficiency 2) Engine-out CO2 emissions compliant to EU zero emission HDV definition 3) Engine-out emissions enabling compliance to relevant emission legislation of targeted application with use of conventional aftertreatment technology			
R&D Plan	Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives Work packages and high-level timeline (indicative 3 years total project): WP1 Requirements and architecture (Months 1 – 2): define KPI's and target values. Define pilot use-cases. Selection of base engine platform. WP2 Engine hardware design (Months 3 – 12): Shipping of base engine to TNO. Identification of required engine adaptations. Design of fuel system, airpath and ignition system architecture together with definition of required component adaptations or updates. Procurement of new parts. WP3 Engine software design (Months 6 – 12): Design of hydrogen engine control software architecture. Development of new engine control algorithms and implementation on development control unit (RCP system). WP4 Engine hardware and software integration (Months 13 – 16): integration of adapted hardware and software components on the base engine realizing the hydrogen engine. WP5 Engine calibration and performance validation (Months 16 – 20): Functional performance verification in TNO test cell. Engine calibration in TNO test cell for use in target application. Performance validation in TNO test cell in standard conditions. Engine performance verification in TNO's Climate chamber for non-standard conditions. WP6 Engine integration into target application (Month 21 – 27): Shipping of hydrogen engine to Korean partner. Integration of the hydrogen engine into the target application. Performance verification. WP7 Pilot demonstration (Month 28 – 36): Providing the target application including the hydrogen			

	engine to an end-user for performance evaluation in a relevant environment. Performance monitoring. WP8 Training and knowledge transfer (Months 1 – 24): throughout the project training and knowledge transfer workshops will be planned. A total of 4 workshops is targeted. An engineer from Korean SME is advised to be on-site at TNO in the Netherlands for significant periods of time.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* TNO (Research Institute) expected roles and responsibilities: - Project management - Specification of hardware and software architecture and component modifications. - Procurement of parts - Integration of updated hardware and software on base engine - Functional verification, calibration and validation of the hydrogen engine at standard and non-standard conditions - Arrange for shipping of Hydrogen engine to Korean SME - Knowledge transfer and training workshops
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* Expected roles and responsibilities of Korean SME partner (anonymous): - Arrange for shipping of base engine to TNO in The Netherlands - Support in interaction between TNO and base engine supplier in case the supplier is a third party - Integration of hydrogen engine into target application accounting for relevant safety aspects and system monitoring - Arrange for pilot demonstration at end-user - Monitoring of performance during pilot

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* Expected outcomes: - Prototype high-efficient hydrogen engine based on existing engine platform - Pilot demonstration of target application including the prototype high-efficient hydrogen engine - Engineers of Korean SME partner are trained on the applied methodologies used during the development pf the high-efficient hydrogen engine Expected impact: - CO2-free power generation contributing to decarbonization of the targeted application sector - CO2-free power generation contributing to decarbonization of the targeted application sector contributing to increased hydrogen demand supporting upscaling of hydrogen availability - Strengthened market position for the Korean SME including high-efficient zero CO2 power generation solution

6. References

(1) Bekdemir, C., Doosje, E., and Seykens, X., “H2-ICE Technology Options of the Present and the Near Future,” SAE Technical Paper 2022-01-0472, 2022, doi:10.4271/2022-01-0472.

(2) X. Seykens, E. Doosje, C. Bekdemir, P. van Gompel, The hydrogen ICE for heavy-duty applications: Towards Ultra-low NOx Emissions, Proc. 43nd International Vienna Motor Symposium, Vienna, April 27 - 29, 2022

(3) X. Seykens, E. Doosje, C. Bekdemir, P. van Gompel, Hydrogen Combustion Concepts: Comparison of Port Fuel Injection with Spark Ignition and High Pressure Direct Injection (HPDITM) – Power Density, Efficiency, and Emissions, Proc. 44th International Vienna Motor Symposium, Vienna, April 26 - 28, 2023

(4) R.C. Willems, X.L.J. Seykens, E. Doosje, C. Bekdemir, P, van Gompel, Optimizing Efficiency and NOx Emissions: Experimental Investigations of Hydrogen High-Pressure Direct Injection on a Heavy-Duty Single-Cylinder Research Engine, Proc. JSAE 2024, Yokohama, May 22 - 24 , 2024

(5) Seykens, X., Doosje, E., Bekdemir, C., and Wezenbeek, P., “Development of Non-Road Spark Ignited H2-ICE with Port Fuel Injection for Fixed Speed Applications,” SAE Technical Paper 2025-01-8435, 2025, doi:10.4271/2025-01-8435.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization